

Stackable Citation Knowledge

Building on Nanopublications for Climate and Biodiversity Research

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A collaboration between VitenHub AS · Knowledge Pixels · Prophet Town

Citation data has two jobs

Credit prior work · works today · Enable building upon · open problem

A reference list entry today

Soroye, P., Newbold, T., & Kerr, J. (2020).
Climate change contributes to widespread
declines among bumble bees across continents.
Science, 367, 685.
doi:10.1126/science.aax8591

→ doi.org/10.1126/science.aax8591

The reader still has to read the body text to learn *which* of Soroye's claims this paper builds on, whether that claim was confirmed, disputed, or qualified, and under which conditions.

A stackable reference list entry

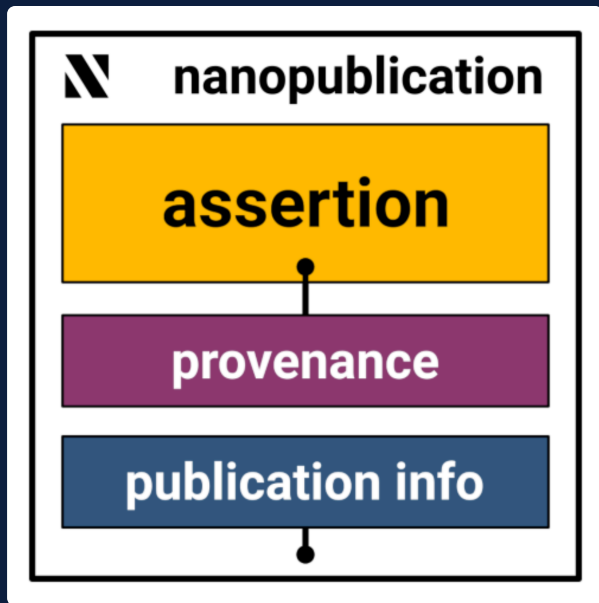
Soroye et al. 2020
cito:qualifies → CiT0 nanopub RAumfa30...
apex Research Synthesis → RA5TJVZ0...
Bombus · Iberia · SSP3-7.0 · 18 nanopubs

→ CiTO: [RAumfa30...](#) · Synthesis: [RA5TJVZ0...](#)

The reader clicks the URI: typed qualification, the sub-claim qualified, the study that produced the qualification, the outcome.

OpenCitations + WikiCite already build the **left**. We add a **citable target** on the right.

Stackable architecture



Why stackable

- Each step in a citation chain is **its own nanopub**
- Each nanopub has **its own URI**
- A future paper can cite **one specific claim**, not just the paper
- CiTO types make the *intent* machine-readable:

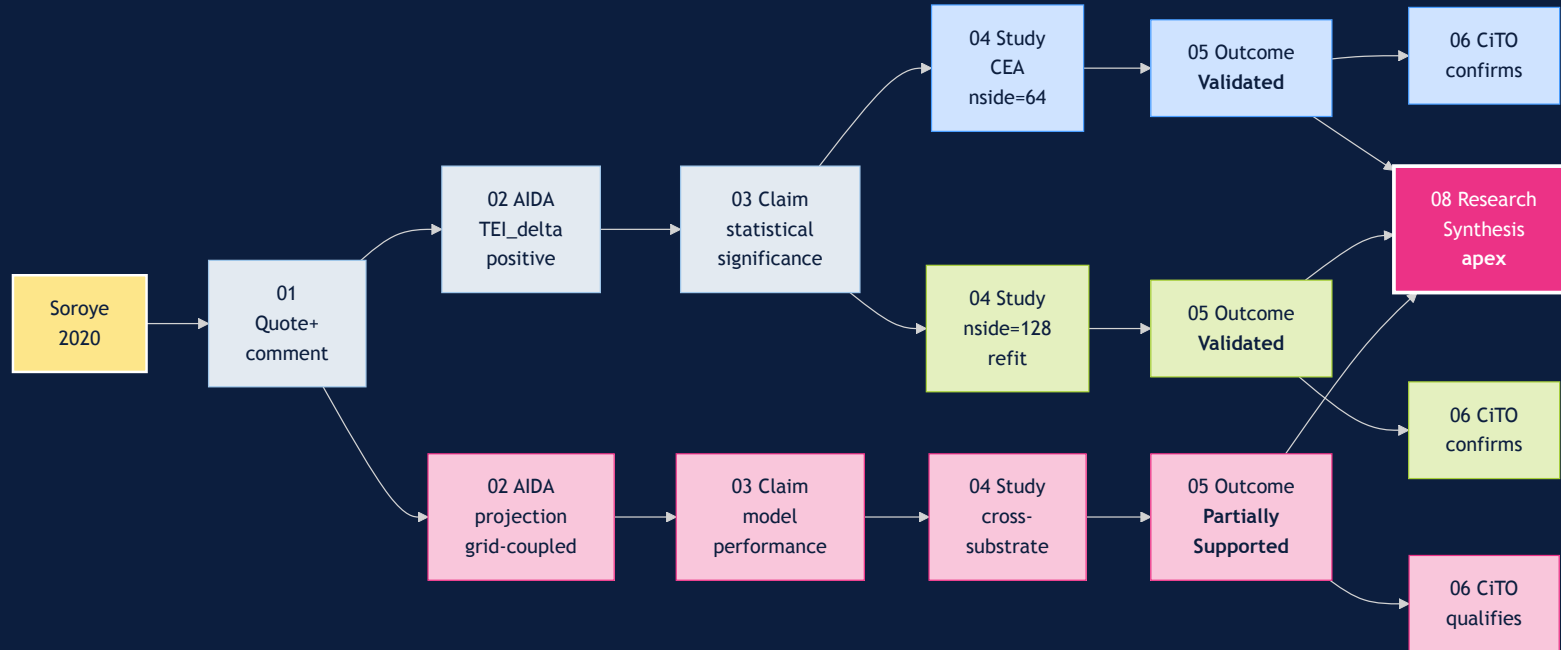
`extends · usesMethodIn · disputes ·
citesAsEvidence · qualifies`

Assertion · RDF triples + Wikidata Q-IDs **Provenance** · PROV-O derivation chain **Publication info** · ORCID · timestamp · CC-BY 4.0

Stackable in practice — a FORRT replication chain

Soroye et al. 2020 (Science) · Iberian *Bombus* under SSP3-7.0 · apex [RA5TJVZ0...]

(https://w3id.org/sciencelive/np/RA5TJVZ0_5Knxzd4OtOoZgO6ZspWHwVCSLWNNd7V9H6QQ) wraps the whole chain into one citable URI



All 16 nodes are addressable nanopubs · see slide 12 for the full clickable URI list.

From form to network

One CiTO citation, end-to-end · Science Live authoring UI on the left · the published nanopub on the right (cito:qualifies Soroye 2020 — RAumfa30...)

Citation with CiTO

Declare citations between papers or other works, using Citation Typing Ontology

Identifier for the citing creative work

https://doi.org/10... or other URL

List citations

Citation Type

extends

DOI (https://doi.org/10...) or other URL of the cited work

https://... or other URL

Invalid URL

+ Add Item

Options

☐ Create Example Nanopub (for testing and demo purposes)

Generate Nanopublication

Template ViewRDF View

Citations

The Creative or Scholarly work

TEI-based extirpation projection is grid-coupled at the projection step (Iberian Bombus, HEALPix nside=64 vs nside=128, SSP3-7.0): mechanism replicates at fit time, per-species ranking does not — fixable

cites the following:

→ extends

https://doi.org/10.5281/zenodo.20113777

→ extends

https://doi.org/10.5281/zenodo.20113780

→ qualifies

Climate change contributes to widespread declines among bumble bees across continents

Author picks the **citation type** (extends · usesMethodIn · disputes · qualifies ...) and the cited DOI · clicks *Generate Nanopublication*

The resulting nanopub on the network · each typed citation is independently click-resolvable

→ Two clicks from "I want to cite this paper" to a signed, addressable, machine-readable citation on the open network.

Three production paths to a signed nanopub

Self-driving cars will drive themselves — but humans still choose the destination. Same for LLM citation extraction: the model does the work · the author chooses the intent.

Extract

Retroactive · automated

Tool stack: GROBID / Anystyle → LLM intent classifier — the work CEC, GRAPHIA, OFFZIB do.

PDF → refs + context
→ LLM → CiTO predicate
→ signed nanopub

Limit: intent recovery has a ceiling.

Author

Prospective · author of citing work

Workflow: open paper in Zotero · pick template · pick CiTO relation · plugin signs + publishes.

Worked example: Soroye 2020 → Bombus chain · Quote-with-comment RAErLL...

Intent at *write time* — no ceiling.

Annotate

Mid-reading · third-party reader

Any expert reader declares the inferred CiTO relation between a paper they're reading and one of *its* references. Signed by the reader's ORCID, contestable, refinable.

Demoed next with Wernberg 2016 and one of its cited works.

Three paths · same destination · same nanopub network · different signers and different epistemic weights.

Three worked domains

One named citable URI per column — these resolve today on the open network

Biodiversity

LifeWatch ERIC

Iberian *Bombus* replication of Soroye
2020 — three branches under SSP3-7.0.

Apex Synthesis [RA5TJVZ0...](#)

Breadth: a PRISMA-compliant
systematic review on quantum-
biodiversity adds **238 more nanopubs**
with full CiTO typing.

Climate change

EarthCARE → DGGS pipeline

Full software stack converting
EarthCARE EO data to HEALPix-geo,
published as a FORRT chain + Research
Software nanopub.

CiTO citation [RAN0HJytFV4...](#)

Cross-discipline

FIESTA — plankton classification

Scattering transforms from
astrophysics (Delouis et al. 2022)
applied to **plankton imagery** (Decrop et
al. 2025) — +8 pp rare-class recall.

One CiTO nanopub, three predicates:

`extends` · `usesMethodIn` ·
`citesAsDataSource`

[RA8BSfq4Cbs3...](#)

On the network today: 18-nanopub *Bombus* chain · 238-nanopub PRISMA review · EarthCARE → DGGS chain · FIESTA cross-disciplinary chain.

This paper *is* 21 nanopubs

7 AIDA cited claims · 6 CiTO citation relations · 8 Wikidata concept definitions

The extended abstract is its own worked example.

Every citation has an inline CiTO type and a click-resolvable URI.

→ Switching to demo

Demo

~4–5 min · pre-recorded

If the recording fails, fallback: open `platform.sciencelive4all.org` in browser and walk through one nanopub URI manually.

Atomicity enables claim-level verification

*Hallucinated and miscited references are increasingly slipping through peer review — even in high-ranked journals. Reviewers are unpaid researchers, and many now use LLMs themselves. "Verify the whole paper" is intractable. **Verify one claim at a time.***

LLM-extracted

lowest

- Pipeline output from intent classifier
- Open to **intent-hallucination**
- *Example:* LLM-classified citation in the PRISMA review

Reader-annotated

mid-low

- Expert reader infers CiTO for *someone else's* citation
- Signed by reader's ORCID, contestable
- *Example:* Wernberg → Poloczanska annotation (demo)

Author-validated

mid-high

- Author of citing work declares intent at write time
- Signed nanopub binds intent to assertion
- *Example:* RAErLL... (slide 6)

FORRT-replicated

highest

- Replication links Outcome to Claim
- Trust becomes a **citation signal**
- *Example:* RAa4QR41... (slide 4)

*Do one thing and do it well — applied to citations. Atomicity makes verification **tractable**, not zero.*

Try it · Cite it · Extend it

Open source · open governance. Your nanopubs live in the decentralised network — not on our servers. Even if Science Live disappears, what you published stays addressable, forever.

Platform



platform.sciencelive4all.org

Open · CC-BY 4.0 · no login

Zotero plugin



ScienceLiveHub/.../zotero

Zotero 7+ · install + walkthrough

Replication scaffold



[ScienceLiveHub/forrt-replication-
template](https://ScienceLiveHub/forrt-replication-template)

Reproducibility · `DOMAIN.md`
adapts

This deck



[ScienceLiveHub/citex2026-
stackable-citations](https://ScienceLiveHub/citex2026-stackable-citations)

Slidev · demo · Zenodo DOI

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References — published nanopubs

Every URI below resolves on the open nanopub network · Science Live's viewer at `platform.sciencelive4all.org/np/?uri=...` serves from its own DB, independent of KP availability

A · Abstract bibliography (9)

Each carries an inline CiTO type.

- Fouilloux 2025 *citesAsDataSource* → [RA-NA9G...](#)
- Groth 2010 *usesMethodIn* → [RAZ1sRo-...](#)
- Groth 2010 *citesAsAuthority* → [RA4h2w5U...](#)
- Kuhn 2021 *usesMethodIn* → [RAgBSZ6a...](#)
- Kuhn 2021 *citesAsPotentialSolution* → [RAh0rL16...](#)
- Page 2021 (PRISMA) *usesMethodIn* → [RAbESId...](#)
- Röseler 2025 (FORRT handbook) *citesAsRecommendedReading* → [RALQHC6a...](#)
- Peroni & Shotton 2012 (CiTO + FaBio) *usesMethodIn* → [RAohoa4...](#)
- Shotton 2010 (CiTO original) *usesMethodIn* → [RAu1zPQ...](#)

B · Bombus chain refs (3)

Cited as text on slides 2 · 4 · 6 · 7 · 10.

- Apex Research Synthesis → [RA5TJVZ0...](#)
- Quote-with-comment (chain step 01) → [RAErLL...](#)
- nside=128 Outcome → [RAa4QR41...](#)

C · Repos referenced

- Talk source: [ScienceLiveHub/citex2026-stackable-citations](#)
- FORRT template: [ScienceLiveHub/forrt-replication-template](#)
- Bombus chain: [annefou/weatherxbiodiversity-projection](#)

D · Platform + API

- Viewer: [platform.sciencelive4all.org](#)
- Query: `/np/constellation?uri=...`